

4.13.14

EE25BTECH11009 - Anshu Kumar Ram

Question:

The perpendicular bisector of the line segment joining $P(1, 4)$ and $Q(k, 3)$ has Y -intercept -4 . Then a possible value of k is

$$(a) \ 1 \quad (b) \ 2 \quad (c) \ -2 \quad (d) \ -4 \quad (0.1)$$

Solution:

Equation of perpendicular bisector:

$$\mathbf{n}^T \mathbf{x} = c \quad (0.2)$$

where direction vector of PQ is

$$\mathbf{PQ} = \begin{pmatrix} k-1 \\ -1 \end{pmatrix}, \quad (0.3)$$

As the perpendicular bisector has normal,

$$\mathbf{n} = \begin{pmatrix} k-1 \\ -1 \end{pmatrix}. \quad (0.4)$$

The midpoint of PQ is

$$\mathbf{M} = \begin{pmatrix} \frac{1+k}{2} \\ \frac{7}{2} \end{pmatrix}. \quad (0.5)$$

And

$$c = -4 \quad (0.6)$$

From (0.3) and (0.6) in (0.2),

$$\begin{pmatrix} k-1 & -1 \end{pmatrix} \begin{pmatrix} -\frac{1+k}{2} \\ -4 - \frac{7}{2} \end{pmatrix} = 0 \quad (0.7)$$

$$\therefore -(k^2 - 1) + 15 = 0 \quad (0.8)$$

$$-k^2 + 16 = 0 \quad \Rightarrow \quad k^2 = 16 \quad (0.9)$$

Hence

$$k = \pm 4 \quad (0.10)$$

From the given options, the valid value is

-4

